

Knapsack Problem

Definition

The **Knapsack Problem** is an optimization problem where we select items with given **weights and values** to maximize the total value without exceeding the bag's capacity.

Algorithm (0/1 Knapsack)

1. Start.
2. Take n items with weights and values.
3. Set the maximum capacity of the knapsack.
4. For each item, check whether it can fit in the knapsack.
5. Choose the item if it gives the maximum value.
6. Do not exceed the maximum capacity.
7. Repeat until all items are checked.
8. Return the maximum value.

Output:

```
90
...Program finished with exit code 0
Press ENTER to exit console. □
```

Time Complexity

For Dynamic Programming 0/1 Knapsack:

$O(n \times W)$

Where:

- n = number of items

- W = knapsack capacity

Space Complexity

$O(n \times W)$

In short:

Knapsack = Select items to get maximum value within a limited weight capacity.